Selective Recovery for Multilingual ASR: Candidate Acquisition and Acceptance

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***Abstract*—We separate fallback acquisition from acceptance using a compression trigger and a fixed four-feature selector. On a locked confirmation panel of 7,500 Common Voice 27.0 (CV27) recordings, selective recovery lowers macro word error rate by 2.26 percentage points (pp) over baseline rescoring and matches unconditional recovery's error counts in every seed. Retrospective CV27 TEST comparisons show 99.98% retention of macro-weighted positive edit gain. Reselection improves unscreened fallbacks; direct adoption can outperform the fixed selector on internally screened HF outputs. Matching acceptance to HF banks with edit-gap weighting yields the lowest exploratory CV27 WER of 31.44%. Across three checkpoints, selective acquisition saves up to 5.15% latency within reranked pipelines. Native HF remains faster but incurs 1.46 pp higher WER on FLEURS and over 2.1 pp on CV27. Code is available at https://github.com/Zhifan-Pan/ISCAS2027.**

***Index Terms*—multilingual speech recognition, selective recovery, candidate acquisition, N-best reranking, confidence estimation, conditional computation**

I. Introduction

Multilingual ASR encounters routine errors and catastrophic repetition loops. Additional decoding helps only when it produces a better candidate that the scoring rule accepts. Separating acquisition from acceptance reveals when recovery helps or wastes computation.

Multilingual N-best reranking [1], [2], discriminative rescoring [3]–[5], voting [6] and confusion networks [7] exploit existing hypotheses. Generative correction expands banks [8]–[11], and risk-based pruning and ASR sampling allocate computation [12], [13]. Conditional computation uses recognizer cascades [14], selective passes [15], encoder proxies [16], selective correction [17] or language-specific distillation [18], but confidence need not predict the benefit of deferral [19].

We distinguish routine beam rescoring from sparse repetition recovery. We ask when bank expansion helps and when reselection helps or harms a screened fallback. The locked panel isolates acquisition; retrospective controls examine acceptance. We compare expected risk, pairwise ranking and minimum Bayes risk on matched banks [20]–[22]. These controls isolate acquisition and acceptance gains within reranked pipelines. Native HF trades higher WER for faster decoding.

II. Controlled Acquisition and Acceptance

A. Recording-level acquisition

We normalize and deduplicate five beams per window i into B_i, with first occurrences and sources retained. With raw top beam b_i and fallback g_i, $F_i = B_i \cup \{g_i\}$ contains at most six candidates. Before recovery, $\mathrm{CR}_{\mathrm{text}}$ is the ratio of top-beam UTF-8 byte length to zlib-compressed length, with zero for empty bytes. The locked gate $z_i = \mathbf{1}\{\mathrm{CR}_{\mathrm{text},i} > 2.4\}$ gives $z^{\mathrm{rec}} = \max_i z_i$. Any trigger expands every window, even singletons (Fig. 1), as a conservative control preserving recording-level invocation. On single-window CV27 audio, $z^{\mathrm{rec}} = z_i$; Section IV-D examines local routing for multi-window FLEURS.

Recovery follows Hugging Face temperatures $0, 0.2, \ldots, 1$ with compression, log-probability and no-speech thresholds 1.35, -1 and 0.6. The internal check compresses serialized token-ID bytes [23]; the outer gate uses decoded UTF-8 text before requesting recovery. Its byte ratio reflects script encoding as well as repetition. Native HF uses the same configuration with conditional retries and returns the final transcript without N-best rescoring.

B. Four-feature shared acceptance

For features $f(y) = (a, p, t, r)$, a is the mean teacher-forced log probability over T body tokens without prefix, padding or end tokens, and $t = \log(1 + T)$. XLM-R-base pseudo-log-likelihood p [24], [25], or PLL, averages individually masked nonspecial subword log probabilities in centered contexts of at most 510 subwords of the original string, rather than within-word masking [26]. Empty outputs have no body tokens or PLL targets and map to $f(\emptyset) = (0, 0, 0, 0)$, but normalization alone need not yield zero features.

For normalized words $w_{1:M}$, the sequence-wide overlapping count is $c_y(u) = \sum_{k=1}^{M-n+1} \mathbf{1}\{w_{k:k+n-1} = u\}$. We measure repetition by

$$r_{n,h}(y) = \frac{\sum_{j=1}^{M-n+1} \mathbf{1}\{c_y(w_{j:j+n-1}) \geq h\}}{\max(1, M - n + 1)}. \tag{1}$$

We use $r(y) = r_{4,3}(y)$, which is zero for $M < 4$. Repetitive sequences also motivate unlikelihood training [27].

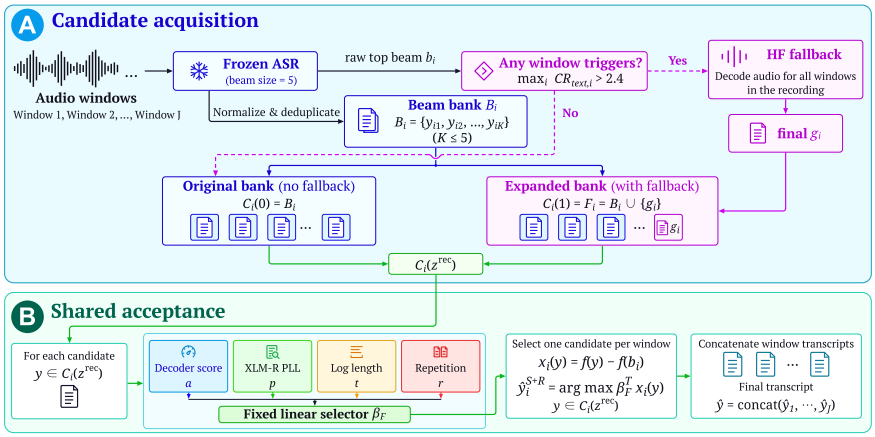


Fig. 1. Acquisition (A) expands every window's candidate bank if any window triggers. Acceptance (B) applies the same fixed four-feature selector to the original or expanded bank. Selected window transcripts are then concatenated.

Centering $x_i(y) = f(y) - f(b_i)$ preserves rankings and softmax probabilities. The empty output thus provides an implicit abstention baseline with score $-\beta_F^\top f(b_i)$. We compute feature-wise root-mean-square (RMS) scales $s \in \mathbb{R}^4_{>0}$ over training-fold pairs $\{(i,k) : i \in \bigcup_{\ell=1}^{L} \mathcal{I}_\ell,\ y_{ik} \in C_i \setminus \{b_i\}\}$ in fitting banks C_i; scales below 10^{-12} are set to one. Define edit benefit $\Delta_{ik} = E(b_i, y_i^*) - E(y_{ik}, y_i^*)$ and $\pi_{ik} = \text{softmax}_k\big(\theta^\top \text{diag}(s)^{-1} x_{ik}\big)$. The softmax and loss cover all $k \in \{1, \ldots, |C_i|\}$, including b_i with $x_i(b_i) = \mathbf{0}$ and $\Delta_{i,b_i} = 0$. Expected-risk fitting [20] minimizes

$$\mathcal{L}(\theta) = -\frac{1}{L}\sum_{\ell=1}^{L} \frac{\sum_{i\in\mathcal{I}_\ell}\sum_{k=1}^{|C_i|}\pi_{ik}\Delta_{ik}}{N_\ell} + \lambda\|\theta\|_2^2. \qquad (2)$$

Here $\mathcal{I}_\ell = \{i : \text{lang}(i) = \ell\}$ indexes fitting windows, N_ℓ counts reference words and $L = 5$.

The shared weights $\beta_F = \text{diag}(s)^{-1}\theta$ are fitted per run on greedy-expanded DEV and fixed across sources, with ties broken by bank order. With $\hat{y}_i^B = \text{argmax}_{y\in B_i} \beta_F^\top x_i(y)$, $C_i(0) = B_i$ and $C_i(1) = F_i$, the policies are

$$\hat{y}_i^{S+R} = \underset{y\in C_i(z^{\text{rec}})}{\text{argmax}}\ \beta_F^\top x_i(y),$$
$$\hat{y}_i^{\text{Direct}} = \begin{cases} \hat{y}_i^B, & z^{\text{rec}} = 0, \\ g_i, & z^{\text{rec}} = 1. \end{cases} \qquad (3)$$

Policy B reranks B_i, and N+R requests and reranks F_i unconditionally. S+R and Direct share acquisition and use B off-trigger, but Direct bypasses selection and PLL on triggers. Window transcripts are concatenated for full-reference scoring.

C. Matched acceptance controls

G4 keeps β_F. On complete HF-expanded DEV banks, H4 refits expected risk over four features and H3 omits PLL with (a, t, r). On these banks, HR4 fits RankNet [21] probabilities $\sigma(\beta^\top(x_{ij} - x_{ik}))$ for $E_j < E_k$, where σ is the logistic function and $x_{ij} = x_i(y_{ij})$. Language-reference normalization and equal total bank weight are retained. HWR4 weights pair losses by $w_{jk} = |E_j - E_k|/c$, where c is the training-fold mean edit gap under the original pair weights.

Within each language, connected clients and reference texts form five folds. Training folds determine RMS scales; held-fold macro WER selects among 49 log-spaced penalties from 10^{-4} to 1, favoring stronger regularization on ties. L-BFGS-B then refits on full DEV from zero. Singletons contribute errors and denominators but no pairs. Fitting and selection exclude padding.

MBR [22] selects $\hat{y}_i = \text{argmin}_{y\in C_i} \sum_{y'\in C_i} q_i(y') E(y, y')$, where E counts word edits. U-MBR sets $q_i(y') \propto 1$ and D-MBR sets $q_i(y') = \text{softmax}(Ta/\tau)_{y'}$. DEV WER selects $\tau \in \{0.1, 0.3, 1, 3, 10\}$ with preference for larger values on ties; candidate ties retain bank order. A *complete policy* uses its acceptor on both banks, whereas *triggered-subset evaluation* retains G4 off-trigger and full-panel denominators.

TABLE I
WER (%) WITH PP CONTRASTS COMPUTED BEFORE ROUNDING. CHECKS DENOTE HF ACQUISITION, CR GATING AND R RESELECTION, WITH † FOR R OFF-TRIGGER.

Path	HF	CR	R	Scorer	FLEURS	CV27
Raw beam-5 top-1	–	–	–	–	18.90	36.41
Native HF [23]	✓	–	–	–	18.55	33.80
B	–	–	✓	G4	17.45	33.98
N+R	✓	–	✓	G4	17.09	31.68
S+R	✓	✓	✓	G4	17.09	31.68
Direct	✓	✓	✓†	G4	17.09	31.46
H3Direct	✓	✓	✓†	H3	18.68	33.64
Reference-assisted TEST oracle bounds						
Original-bank oracle					16.19	30.19
Full-HF oracle					15.83	27.68
CR-routed oracle					15.83	27.68
S+R gap to routed oracle, pp					1.26	4.00

Locked CV27 confirmation panel (7,500 recordings)

	B	N+R = S+R	Gain
Macro-average	35.02	32.75	2.26
Contrast	ΔWER		95% upper bound
S+R − B	-2.26		-1.32
S+R − N+R	0.00		0.00

III. Data and Evaluation

We adapt Whisper-small [29] on 44.60 h of FLEURS TRAIN [28] across Arabic, Persian, Polish, Turkish and Italian using LoRA [30] over three random seeds (4,656 updates, rank and scale four, zero dropout across all attention projections). LoRA training excludes CV27 TRAIN, and selectors and

gates are tuned on CV27 DEV. Evaluation on FLEURS TEST (13.07 h; 3,665 recordings) and CV27 TEST (83.00 h; 58,514 recordings) [31] is retrospective because both splits were accessed during development.

The locked confirmation panel contains 7,500 CV27 TRAIN recordings, 1,500 per language, with no previously accessed recordings, audio or reference texts. Predictions, stopping rules and the one-sided 0.10 pp noninferiority margin were locked before reference evaluation under procedural controls. We account for within-panel correlation with 10,000 synchronized bootstrap draws of speaker components within languages. Recurring speakers across splits restrict generalization to this speaker-clustered sampling frame. Confirmatory evidence covers the original G4 acquisition policy; later gates, refits and window policies remain retrospective.

We estimate costs from 255 FLEURS and 472 CV27 recordings using inverse sampling probabilities. The sample includes all first-checkpoint triggers or Native HF output changes plus ten ordinary recordings per language and duration quintile. Study A measures four paths on three checkpoints; Study B measures seven on one. Five FLEURS and 207 CV27 later-checkpoint triggers are unsampled (0.05% and 0.12% of seed–recordings). Triggered N+R and S+R execute identical acquisition and scoring. Study B covers every trigger on its checkpoint and corroborates Study A's savings (5.18% and 4.54% versus 5.15% and 4.57%), though unobserved timing variation is not bounded. Both use three randomized paired rounds on a server with four NVIDIA A40 GPUs (48 GB each), with resident models, within-path encoder and first-search reuse, and 32-mask PLL batches. Waveform-to-output timing excludes loading, warmup and interfered blocks, and peak CUDA memory in MiB is the maximum whole-call allocation across recordings and rounds. Session-conditional intervals use 4,000 paired stratified draws and finite-population correction; Study A applies equal checkpoint weights and synchronized draws.

Decoding uses known languages, bfloat16, five beams, length penalty one, native suppression and a 444-token cap in Transformers 4.57.6. At 16 kHz, N samples give $J = \lceil N/480000 \rceil$ windows bounded by $\lfloor jN/J \rfloor$ without previous-window text conditioning. Deduplication, WER and r share NFC normalization, digit unification, punctuation-to-space, Persian mapping and Turkish casing [32], with fixed source RNG mappings. WER aggregates within languages before averaging five languages and three seeds, and calls pool seed–recordings. TEST intervals use 2,000 paired draws of client/reference components within languages, synchronized across seeds and conditional on frozen fits and source draws. FLEURS comparisons are descriptive because component metadata are unverified.

IV. Results

A. Acquisition with fixed acceptance

On the locked confirmation panel, S+R gains 2.26 pp over B, with seed gains of 0.20 to 4.54 pp (Table I). Matching N+R's error counts in every seed meets the prespecified 0.10 pp margin but does not guarantee zero loss under distribution shift. Of 61 triggers, 49 improve edits and two worsen them.

TABLE II
Same-HF TEST WER (%) and gain B−S (pp) before rounding. Checks denote PLL, pairwise P and edit-gap weighting W.

Acceptor	PLL	P	W	FLEURS			CV27		
				B	S	Gain	B	S	Gain
G4 [20]	✓	–	–	17.45	17.09	0.37	33.98	31.68	2.30
H4 [20]	✓	–	–	17.46	17.09	0.37	33.97	31.52	2.45
HR4 [21]	✓	✓	–	17.51	17.14	0.37	34.02	32.30	1.72
H3 [20]	–	–	–	19.05	18.68	0.37	36.15	33.77	2.38
HWR4 [21]	✓	✓	✓	17.47	17.10	0.37	33.97	31.44	2.53
U-MBR [22]	–	–	–	19.14	19.14	0.00	36.91	36.88	0.03
D-MBR [22]	–	–	–	18.92	18.75	0.17	36.09	34.48	1.60
A. Original family: CV27 ΔWER (individual 97.5% CI)									
Triggered HR4 − H4				+0.72			[+0.40, +1.16]		
Complete H3 − H4				+2.25			[+2.16, +2.34]		
B. Later two-comparison family									
Triggered U-MBR − H4				+2.51			[+1.93, +3.18]		
Triggered HWR4 − H4				-0.08			[-0.15, -0.03]		

Retrospective evaluation yields 32.71% WER for Direct versus S+R's locked 32.75%. Both beat B and Native HF's retrospective 34.87% by over 2.1 pp. Direct's unrounded difference from S+R is -0.05 pp with a paired speaker-bootstrap 95% interval from -0.19 to $+0.04$ pp.

On TEST, S+R retains all ten beneficial FLEURS instances at 0.09% calls and 370 of 373 CV27 instances at 0.30%, with five edits missed. Retention is 99.98% of available positive edit gain after language-reference normalization and averaging over languages and seeds. Normalized outputs differ on one FLEURS and 13 CV27 nontriggers despite nearly identical WER. On these nontriggers, S+R keeps B, but N+R requests g_i and reselects from the expanded bank. Reference-assisted oracles minimize recording edits over concatenated original, expanded or CR-routed banks. Routing changes this bound by under 0.01 pp versus CV27 S+R's 4.00 pp gap, so acceptance offers greater headroom.

B. Interaction with the fallback source

S+R can reject unscreened fallbacks and beats Direct by 0.53 pp with greedy recovery and 0.43 pp with one $T = 0.2$ draw. HF screens retries by compression, log-probability and no-speech criteria. The greedy-fitted G4 can instead favor an original beam over a screened recovery. Direct then leads by 0.22 pp, with a 95% interval from 0.11 to 0.37 pp. The gain comes from one checkpoint (-0.68 pp versus $+0.00$ and $+0.02$ pp). DEV favored HF reselection at 30.08% versus 30.09% and guided the fixed acquisition control.

Direct retains 67.13% of positive gain from unconditional final HF, compared with S+R's 99.98% from N+R. Relative to its own control, Direct forgoes 1.24 pp of benefit but avoids 3.58 pp of harm.

C. Acceptance on matched HF banks

After refitting on HF banks, H4 improves over G4 but trails Direct (31.52% versus 31.46% CV27 WER). Weighted HWR4 yields the lowest complete CV27 WER of 31.44%

TABLE III
POPULATION-WEIGHTED MS AND OBSERVED PEAK ALLOCATED MIB WITH BOTH MODELS RESIDENT. EACH STUDY USES THREE PAIRED ROUNDS.

Path	FLEURS		CV27	
	ms	MiB	ms	MiB
A. Four-path comparison across three checkpoints				
Native HF	1291.7	2980.8	517.4	2813.8
B	1352.7	3144.9	591.9	3133.1
N+R	1436.8	2988.3	639.2	3133.1
S+R	1362.8	3189.7	610.0	3133.1
B. Seven-path comparison on one checkpoint				
Native HF	1304.8	2978.4	531.2	2811.8
Raw top-1	1222.6	2946.6	479.9	2571.0
B	1364.0	3142.4	605.1	3131.1
N+R	1449.6	2986.3	657.9	3131.1
S+R	1374.5	3187.2	628.0	3131.1
Direct	1372.8	3187.2	625.8	2811.8
H3Direct	1314.0	3187.2	586.4	2811.8
Relative latency saving: S+R versus N+R (95% CI)				
Study A	5.15% [2.84, 9.41]		4.57% [4.40, 4.73]	
Study B	5.18% [2.88, 9.39]		4.54% [4.44, 4.66]	
Matched-study B-path increments				
B − Raw top-1, ms	141.4		125.2	
N+R − B, ms	85.6		52.8	
S+R − B, ms	10.5		22.9	
Fallback increment saved	87.7%		56.6%	

(Table II, where S denotes each acceptor's selective policy). Triggered HR4 trails H4 in every seed ($+0.91$, $+0.08$ and $+1.17$ pp). Edit-gap weighting reverses the mean contrast to -0.08 pp through a 0.25 pp gain in one checkpoint. Intervals provide 95% Bonferroni coverage per family, with 10,000 synchronized draws for the later family.

FLEURS gains round to 0.37 pp for fitted acceptors because S equals its own B off-trigger. H3's B has two fewer errors on one of ten triggers, but S errors match and the average difference is below 0.006 pp. Nontriggers account for 97.23% of the refitted H3–H4 WER gap in routine selection, without isolating PLL's causal contribution or time share.

D. Cost, heterogeneity and supporting diagnostics

S+R preserves N+R's accuracy gain at lower overhead for systems using B (Table III). Study B measures 141.4 ms and 125.2 ms for rescoring above raw top-1 on FLEURS and CV27. S+R saves 87.7% and 56.6% of N+R's additional recovery cost. An unweighted 25-recording CV27 profile averages teacher scoring at 85.9 ms and PLL at 149.3 ms, including two singletons that skip both. These conditional costs motivate selective PLL, but the larger population singleton share prevents a direct stage decomposition. Native HF skips N-best rescoring and is faster at a WER cost of 1.46 pp on FLEURS and over 2.1 pp on CV27 TEST.

H3Direct uses H3 off-trigger and final HF on triggers, and omits PLL like raw top-1. On CV27 it saves 6.29% over Direct but trails Native HF by 55.2 ms, with a 95% interval of 52.3 to 58.2 ms; the FLEURS interval spans zero. H4 and HWR4 only replace G4's weights; selective PLL remains future work. In Study B, Direct bypasses scoring on the CV27 trigger responsible for B's peak CUDA memory. On FLEURS, Direct and H3Direct match S+R's peak CUDA allocation in two-window recovery with both windows' state retained.

TABLE IV
TEST GAINS AVERAGED OVER SEEDS. CALLS POOL SEED–RECORDINGS.

Language	FLEURS		CV27	
	Gain, pp	Calls, %	Gain, pp	Calls, %
Arabic	0.64	0.23	1.40	0.11
Persian	1.20	0.27	9.05	1.15
Polish	0.00	0.00	0.30	0.33
Turkish	0.00	0.00	0.51	0.03
Italian	0.00	0.00	0.23	0.02
Empty-HF acceptance diagnostic (18 instances, 11 recordings)				
Acceptance minus B reversion	$S = -118$	$D = +120$	$I = -1995$	

Only HF introduces empty TEST candidates. Under signs $(+, +, -, -)$, the centered abstention baseline outranks candidates with $a, p \leq 0$, $t > 0$ and $r \geq 0$, even without repetition. Accepting 18 empties in 11 CV27 recordings removes 1,995 insertions, with S/D/I tradeoffs in Table IV. This accords with hallucination suppression [33], although silence labels are needed to establish acoustic causes. Reverting these acceptances to B retains 91.52% of gain, or 2.10 pp at 31.88% WER. Refitting H4 with $e(y) = \mathbf{1}\{y = \emptyset\}$ on the same DEV folds and grid retains all 18 acceptances at 31.52% WER, but only one or two DEV empties per checkpoint limit calibration.

Raw top-1 to B measures routine rescoring; B to S+R isolates sparse recovery. Persian contributes 78.77% of CV27 TEST gain and 84.07% of gain on the locked confirmation panel (Table IV). Excluding Persian, the four-language macro gain remains 0.61 pp on CV27 TEST and 0.45 pp on the locked confirmation panel. Polish, Turkish and Italian have no FLEURS triggers or gains. Script encoding, acoustics and low-rank adaptation may interact, but their effects are not isolated. The 22 two-window FLEURS recordings yield 52.69% of net gain. On cached TEST banks, local S+R routing reduces window requests from 15 to 10 with unchanged error counts. Local Direct changes WER by under 0.01 pp, although live request order may alter RNG draws.

V. CONCLUSION

Selective acquisition matches unconditional recovery across seeds on the locked confirmation panel. It separates sparse recovery from routine rescoring to reduce recovery overhead. Retrospective controls favor expected risk over unweighted ranking on matched HF banks in every seed.

Acceptance should match fallback screening. Reselection filters unscreened draws; Direct avoids rescoring internally screened HF outputs. Source-matched HWR4 yields the lowest exploratory CV27 WER, but acceptance gains depend on checkpoint. Window-local routing and confidence-gated PLL could further reduce acquisition and routine rescoring costs.

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